Cell energy: mitochondria, ATP and cellular respiration

1. Why cells need energy

Every living cell does work all the time. It builds proteins, copies DNA, moves molecules across its membrane, and in muscle cells it contracts. All of this work needs energy. Cells get that energy from food, mostly from glucose, a simple sugar with six carbon atoms. But a cell cannot spend glucose directly, the same way you cannot pay for a jeepney ride with a sack of rice. The energy in glucose first has to be moved into a smaller, easy-to-spend form.

2. ATP: the energy currency of the cell

ATP stands for adenosine triphosphate. It is the molecule cells use to store and spend energy for their daily work. An ATP molecule has three parts: the base adenine, the sugar ribose, and a chain of three phosphate groups. The bonds between the phosphate groups hold a lot of energy. When the cell needs energy, an enzyme breaks off the last phosphate group. This turns ATP into ADP (adenosine diphosphate) and a free phosphate, and it releases energy the cell can use right away.

The cell then recharges ADP back into ATP by adding a phosphate again, using energy from food. This is the ATP-ADP cycle. A good way to remember it: ATP is a charged battery and ADP is a used battery. A single working muscle cell can use and rebuild about ten million ATP molecules every second, so the cycle never stops.

3. Key terms for this page

Glucose: the six-carbon sugar cells break down for energy. ATP: adenosine triphosphate, the cell's energy currency, with three phosphate groups. ADP: adenosine diphosphate, ATP after it has lost one phosphate group. Enzyme: a protein that speeds up a chemical reaction in the cell.

4. Mitochondria: the powerhouse of the cell

Mitochondria (singular: mitochondrion) are the organelles where most of the cell's ATP is made. They are bean-shaped and have two membranes. The outer membrane is smooth. The inner membrane is folded into many ridges called cristae, which give it a much bigger surface area. The fluid inside the inner membrane is called the matrix. Cells that use a lot of energy, like muscle cells and nerve cells, have many more mitochondria than cells that use little. Mitochondria also have their own small loop of DNA, which is passed down from the mother.

5. Cellular respiration: the big picture

Cellular respiration is the process cells use to release the energy in glucose and store it in ATP. It needs oxygen, which is why it is called aerobic respiration. The overall equation is:

glucose + oxygen -> carbon dioxide + water + energy (ATP), or C6H12O6 + 6 O2 -> 6 CO2 + 6 H2O + about 36 to 38 ATP.

The carbon dioxide you breathe out is a waste product of this process, and the oxygen you breathe in is used at its very last step. Respiration happens in three stages, one after the other: glycolysis, the Krebs cycle, and the electron transport chain.

6. The three stages of cellular respiration

Stage 1, glycolysis, happens in the cytoplasm, outside the mitochondria. One glucose molecule is split into two molecules of pyruvate. It does not need oxygen and makes a net gain of 2 ATP. Stage 2, the Krebs cycle (also called the citric acid cycle), happens in the matrix of the mitochondria. Pyruvate is broken down completely, releasing carbon dioxide. It makes 2 ATP and loads energy-carrying molecules called NADH and FADH2. Stage 3, the electron transport chain, happens on the inner membrane of the mitochondria (the cristae). NADH and FADH2 pass their electrons down a chain of proteins, and oxygen accepts the electrons at the end, forming water. This stage makes most of the ATP: about 32 to 34 molecules.

7. When there is no oxygen: fermentation

If oxygen runs out, cells can still run glycolysis, but the later stages stop. To keep glycolysis going, cells use fermentation. In human muscle cells, pyruvate becomes lactic acid; this is lactic acid fermentation, which happens during hard exercise. Yeast turns pyruvate into ethanol and carbon dioxide; this is alcoholic fermentation, used to make bread and tuba. Fermentation makes only the 2 ATP from glycolysis, much less than aerobic respiration.

8. Review questions

Where in the cell does glycolysis happen? Which stage makes the most ATP? What is the role of oxygen in the electron transport chain? Why do muscles feel tired after a long sprint?